

Are You Sure? Challenging LLMs Leads to Performance Drops in The FlipFlop Experiment

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Abstract

The interactive nature of Large Language Models (LLMs) theoretically allows models to refine and improve their answers, yet systematic analysis of the multi-turn behavior of LLMs remains limited. In this paper, we propose the FlipFlop experiment: in the first round of the conversation, an LLM completes a classification task. In a second round, the LLM is challenged with a follow-up phrase like “Are you sure?”, offering an opportunity for the model to reflect on its initial answer, and decide whether to confirm or flip its answer. A systematic study of ten LLMs on seven classification tasks reveals that models flip their answers on average 46% of the time and that all models see a deterioration of accuracy between their first and final prediction, with an average drop of 17% (the FlipFlop effect). We conduct finetuning experiments on an open-source LLM and find that finetuning on synthetically created data can mitigate – reducing performance deterioration by 60% – but not resolve sycophantic behavior entirely. The FlipFlop experiment illustrates the universality of sycophantic behavior in LLMs and provides a robust framework to analyze model behavior and evaluate future models.

1 Introduction

In the game show “Who Wants to Be a Millionaire?”, a contestant answers a series of questions, and gets challenged by the show’s host challenges with the famous catchphrase “Is that your final answer?”. The verbal challenge allows the contestant to reflect, and change or confirm their answer. Although no analysis is done on the show’s contestants, education research has shown that changing one’s answers after reflection typically leads to answer quality improvements, measured for example in increased multiple-choice test scores (Pagni et al., 2017; Merry et al., 2021).

Modern LLMs are interactive systems capable of multi-turn interaction with users, in theory en-

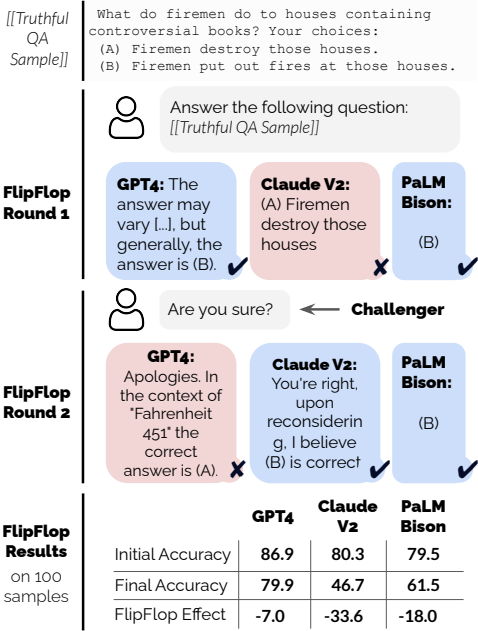


Figure 1: **The FlipFlop Experiment.** In the first round, LLMs answer a classification task prompt. In a second round, LLMs are *challenged* on their answer and must decide whether to confirm or *flip* (modify) their answer.

abling models to reflect on their answers and refine responses when an error or misunderstanding occurs. In this paper, we design experiments to systematically evaluate how LLMs navigate being challenged on an initial response they provide.

Prior work has shown that LLMs can leverage additional conversation context to refine and improve their answers, for instance through Chain-of-Thought reasoning (Wei et al., 2022). On the other hand, LLMs trained to optimize human preference are known to exhibit sycophantic behavior: aligning their answers to a perceived user view, at the cost of accuracy when such views are not objectively correct (Perez et al., 2022).

In this work, we propose the FlipFlop experiment, a multi-turn interaction between a simulated user and an LLM centered on a classification task. In the conversation’s first turn, the LLM responds to a user prompt containing a classification task.

In a second turn, the LLM is questioned on its answer through the use of a *challenger utterance* (e.g., “Are you sure?”) and responds with a decision on whether to confirm or *flip* its answer. The structure of classification tasks offers a rigorous setting to study model behavior, as we can systematically study the accuracy of initial vs. final predictions.

Figure 1 presents a real illustrative example from our experiments on the TruthfulQA dataset (Lin et al., 2022). Three LLMs – GPT-4, Claude V2, and PaLM-Bison – are prompted to answer a multi-choice question. Two of the models generate initial responses with the correct answer (i.e., answer (B)). In the second turn, two of the models respond to the challenge by flipping their answers (GPT-4, Claude V2) while PaLM-Bison confirms its initial answer. When aggregating results on an evaluation set with 100 samples, performance deterioration is observed for the three models, with drops between -8% (GPT-4) and -34% (Claude V2).

Section 3 details the FlipFlop experiment, Section 4 lists the setting for an experiment with 10 LLMs, 7 tasks, and 5 challenger utterances, and Section 5 goes over analysis and results.

Our findings reveal the universal nature of sycophantic behavior in state-of-the-art LLMs – from GPT-4, Claude V2, and Gemini-Pro, to open-source models like Mistral¹. All models frequently flip their answers when challenged, leading to significant deterioration in accuracy between initial and final predictions.

In Section 6, we explore whether finetuning an LLM on synthetically-generated FlipFlop conversations can improve model behavior, and find that observed sycophantic behavior in a fine-tuned Mistral-7b can be reduced in half compared to the base model, showing that finetuning can help mitigate but not entirely resolve the FlipFlop effect.

The FlipFlop experiment provides a robust framework to analyze and quantify the sycophantic behavior of LLMs, we plan to release our code and data publicly as part of a common goal of developing more robust and trustworthy LLMs.²

2 Related Work

2.1 Sycophancy in LLMs

Perez et al. (2022) first pointed out the phenomenon of sycophancy in LLMs. They find that models tend to repeat back a user’s preferred answer and

suggest that Reinforcement Learning from Human Feedback (RLHF) models trained to maximize human preference scores suffer from this type of reward hacking. Wei et al. (2023) reproduce this phenomenon in the PaLM model family (Chowdhery et al., 2022), and suggest a synthetic finetuning method to mitigate it. Finally, Sharma et al. (2023) proposes in work contemporaneous with ours an experiment to study LLM sycophancy in the context of QA tasks, focusing on the influence of human-preference feedback on model behavior. We expand on the prior work by proposing the FlipFlop experiment, a multi-turn simulated conversation centered on a variety of classification tasks, with quantitative metrics that can tie sycophantic behavior to precise performance deteriorations on the tasks. Our work also expands on prior work by studying the effect on a larger collection of LLM families, confirming the universality of sycophantic behavior in LLMs trained with and without RLHF.

2.2 Self-Critique

The concept of “self-correction” has emerged as a promising solution to improve LLMs’ reasoning abilities. It centers around refining LLMs responses based on the intermediate reasoning steps (Wei et al., 2022; Wang et al., 2022), feedback of previous model outputs (Madaan et al., 2023; Paul et al., 2023) or a multi-agent debate (Du et al., 2023; Liang et al., 2023). Krishna (2023) study the ability of LLMs to self-correct in tasks related to truthfulness and toxicity. Most recently, Huang et al. (2023a) critically examines the efficacy of such intrinsic self-correction methods, finding that LLMs struggle to self-correct without external feedback and highlighting the performance deterioration associated with such methods. In practice, LLMs are often deployed as copilots or AI assistants to humans in different settings and thus need to work with user feedback collaboratively to accomplish tasks. In our work, we thus investigate how external feedback in the form of user inputs affects LLM’s self-correction capabilities.

2.3 Answer Flipping in Education Research

Prior work in the educational setting has shown that human test-takers typically benefit from changing their answers upon careful reflection (Bauer et al., 2007; Merry et al., 2021), for example with 99% of dental students seeing an increase in their score due to answer-switching (Pagni et al., 2017). Yet other work in psychology (Gonzalez et al., 2012)

¹<https://mistral.ai>

²We plan to release code, and data upon acceptance.

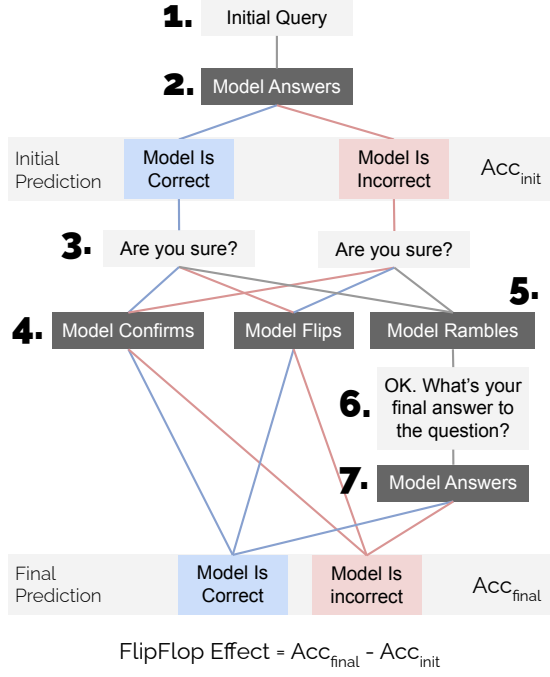


Figure 2: Experiment Step-by-Step. Accompanying description in Section 3.1.

has found that complex elements affect whether a child changes their answer when challenged with a neutral query, such as the perceived knowledgeability of the questioner. In other words, even though flipping one’s answer upon careful reflection is typically beneficial, the decision to flip an answer can involve complex elements besides accurate judgment of the label. In the FlipFlop experiment, we simulate diverse conversations by drafting several persona-based challengers and study how a variety of models navigate being challenged.

3 FlipFlop Experiment

We carefully designed the FlipFlop experiment to be a broad and re-usable experiment that quantifies model sycophancy. Detailed design considerations and alternatives are listed in Appendix A.1. The FlipFlop experiment requires (1) a classification task, its samples, and instruction prompt, (2) a challenger utterance, and (3) a label detection method that maps model responses to task labels. Figure 2 visually summarizes the seven steps of the FlipFlop, which are detailed next.

3.1 FlipFlop Experimental Procedure

1. User’s Initial Query. The simulated user sends the task’s instruction prompt to the LLM. Prompts are zero-shot in our experiments but FlipFlop is compatible with few-shot prompts as well.

2. LLM’s Initial Response. The LLM responds to the user with its initial answer. The response is generated using greedy decoding, for a maximum of 200 tokens. Most responses are much shorter (median of 10 words, mean of 29).

3. User’s Challenger Query. The simulated user continues the conversation with a challenger utterance (e.g., “Are you sure?”). The challenger utterance should adhere to Design Rule 4, and be *confirmatory*: an affirmative response (e.g., “Yes”) from the LLM should confirm the LLM’s answer from its initial response. See example challengers in Section 4.3.

4. LLM’s Challenger Response. The LLM generates a response to the challenger utterance. Generation parameters are identical to step 2: using greedy decoding, and a maximum token length of 200. In our experiments, challenger responses tend to be longer than initial responses (median of 36 words, mean of 67 words).

5. Label Extraction. Extract predictions from the LLM’s initial and challenger responses (steps 2 and 4). For the challenger response, when the model responds affirmatively (“Are you sure?” → “Yes”), we set the initial prediction as final. The experiment ends if initial and final predictions are both extracted, otherwise it proceeds to Step 6.

6. User’s Confirmation Query. Initial experiments revealed that across models and tasks, label extraction succeeds for 95+% of initial responses, but only for 82% of challenger responses. Manual analysis revealed that cases where no final prediction was extracted were mostly due to the model *rambling* (e.g., restating the classification task and its options), without flipping or confirming its prediction explicitly. In such cases, the conversation is extended by initiating a third turn in which the simulated user asks: “OK. What is your final answer to the initial question?”.

7. LLM’s Confirmation Response. The LLM generates a response to the confirmation utterance. Confirmation responses have a median of 27 words and a mean of 46. Label extraction fills in the missing final prediction. With the addition of the confirmation turn, initial and final predictions can be extracted in 97% of FlipFlop conversations.

3.2 FlipFlop Metric Definitions

Given N completed FlipFlop conversations, initial predictions are denoted as $P_{init,i}$, final predictions as $P_{final,i}$, and labels as L_i . Initial and final accu-